

Numerical Differentiation Master Sheet

Section 4.1 & Lecture 22 Reference

1 General Theory

Usage: Use this to derive any specific formula by using Lagrange polynomials. It represents the slope of the interpolating polynomial at a specific point.

$$f'(x_j) = \sum_{k=0}^n f(x_k) L'_{n,k}(x_j) + \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0, i \neq j}^n (x_j - x_i)$$

2 Two-Point Formulas ($O(h)$)

Usage: Use when data is very limited. Forward is for the first point; Backward is for the last.

- **Forward:** $f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h} - \frac{h}{2} f''(\xi)$
- **Backward:** $f'(x_0) = \frac{f(x_0) - f(x_0-h)}{h} + \frac{h}{2} f''(\xi)$

Derivation of Optimal h :

1. **Total Error:** $E(h) = \text{Round-off} + \text{Truncation}$
2. $E(h) = \frac{2\epsilon}{h} + \frac{h}{2} M_2$
3. **Differentiate:** $E'(h) = -\frac{2\epsilon}{h^2} + \frac{M_2}{2}$
4. **Set to zero:** $0 = -\frac{2\epsilon}{h^2} + \frac{M_2}{2}$
5. $h^2 = \frac{4\epsilon}{M_2} \implies h = \sqrt{\frac{4\epsilon}{M_2}}$

3 Three-Point Midpoint ($O(h^2)$)

Usage: Use for any point where you have one neighbor on both sides ($x_0 - h, x_0 + h$). Most stable standard formula.

$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi)$$

Derivation of Optimal h :

1. $E(h) = \frac{\epsilon}{h} + \frac{h^2}{6} M_3$
(RE is $\frac{2\epsilon}{2h} = \frac{\epsilon}{h}$)
2. $E'(h) = -\frac{\epsilon}{h^2} + \frac{2hM_3}{6} = -\frac{\epsilon}{h^2} + \frac{hM_3}{3}$
3. **Set to zero:** $\frac{\epsilon}{h^2} = \frac{hM_3}{3}$
4. $h^3 = \frac{3\epsilon}{M_3} \implies h = \sqrt[3]{\frac{3\epsilon}{M_3}}$

4 Three-Point Endpoints ($O(h^2)$)

Usage: Use at table edges ($x_0, x_0 + h, x_0 + 2h$ for start; $x_0, x_0 - h, x_0 - 2h$ for end).

- **Forward:** $\frac{1}{2h}[-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$
- **Backward:** $\frac{1}{2h}[3f(x_0) - 4f(x_0-h) + f(x_0-2h)]$

Error Term: $\frac{h^2}{3} f^{(3)}(\xi)$

Derivation of Optimal h :

1. $E(h) = \frac{4\epsilon}{h} + \frac{h^2}{3} M_3$
(RE is $\frac{(3+4+1)\epsilon}{2h} = \frac{4\epsilon}{h}$)
2. $E'(h) = -\frac{4\epsilon}{h^2} + \frac{2hM_3}{3}$
3. **Set to zero:** $\frac{4\epsilon}{h^2} = \frac{2hM_3}{3}$
4. $12\epsilon = 2h^3 M_3 \implies h^3 = \frac{6\epsilon}{M_3}$
5. **Result:** $h = \sqrt[3]{\frac{6\epsilon}{M_3}}$

5 Five-Point Midpoint ($O(h^4)$)

Usage: Use for maximum precision if you have two points on each side ($x_0 \pm h, x_0 \pm 2h$).

$$f'(x_0) = \frac{f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)}{12h}$$

Error Term: $\frac{h^4}{30} f^{(5)}(\xi)$

Derivation of Optimal h :

1. $E(h) = \frac{1.5\epsilon}{h} + \frac{h^4}{30} M_5$
(RE is $\frac{(1+8+8+1)\epsilon}{12h} = \frac{1.5\epsilon}{h}$)
2. $E'(h) = -\frac{1.5\epsilon}{h^2} + \frac{4h^3 M_5}{30}$
3. **Set to zero:** $\frac{1.5\epsilon}{h^2} = \frac{2h^3 M_5}{15}$
4. $22.5\epsilon = 2h^5 M_5 \implies h^5 = \frac{11.25\epsilon}{M_5}$
5. **Result:** $h = \sqrt[5]{\frac{11.25\epsilon}{M_5}}$

6 Five-Point Endpoint ($O(h^4)$)

Usage: Highest accuracy for edges. Requires 4 consecutive neighbors.

$$f'(x_0) = \frac{-25f_0 + 48f_1 - 36f_2 + 16f_3 - 3f_4}{12h}$$

Error Term: $\frac{h^4}{5} f^{(5)}(\xi)$

Derivation of Optimal h :

1. $E(h) = \frac{10.67\epsilon}{h} + \frac{h^4}{5} M_5$
2. $E'(h) = -\frac{10.67\epsilon}{h^2} + \frac{4h^3 M_5}{5}$
3. $0.8h^5 M_5 = 10.67\epsilon$
4. **Result:** $h = \sqrt[5]{\frac{13.34\epsilon}{M_5}}$

7 Critical Definitions

- **Truncation Error:** The error from using a finite formula instead of a limit. Decreases as $h \rightarrow 0$.
- **Round-off Error:** The error from computer precision. Increases as $h \rightarrow 0$.
- **M:** Max value of the $(n+1)$ -th derivative on the interval $[a, b]$.

8 Exam Strategy

1. **Find Target x :** If it's in the center, use **Midpoint**. If it's at the start or end, use **Endpoint**.
2. **Count Available Points:** If you have 5 points, use the $O(h^4)$ formula for significantly higher accuracy.
3. **Optimal h Calculation:** Calculate theoretical h using the root formulas above, then select the h from the table that is closest to your calculated value.